Wind Turbine Project

University of California, Mechanical Engineering Dept.

ENGIN 26

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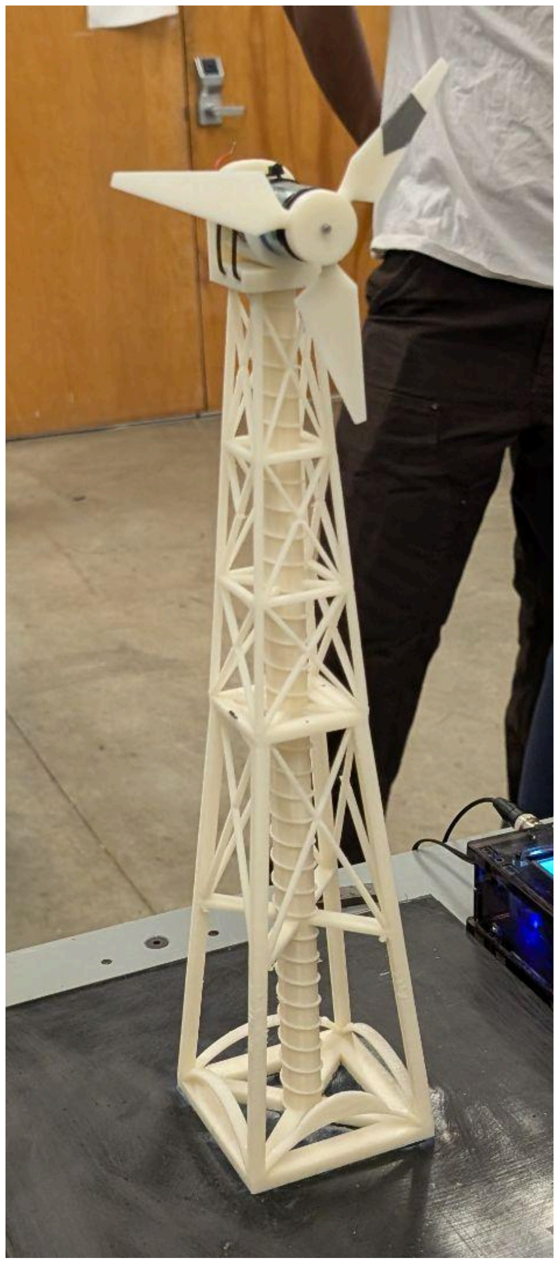


Project Summary

The main purpose of this project is to design a wind turbine that would be able to produce electricity under laboratory test conditions at a wind speed of 25 miles per hour. It was necessary to validate all major characteristics of the wind turbine with research, verify the mechanical strength with FEA, manufacture with 3D printing, and test both mechanical strength characteristics of the turbine tower and its ability to produce electrical energy.

The final turbine incorporates a three-bladed rotor coupled to a tapered square truss tower (Figure 1). The rotor features a NACA 4412 blade profile chosen for its high lift-to-drag ratio in low-speed winds, with an angle of attack of 6° and blade twisting from 20° to 0° from the root to the tip. The tower itself is constructed as an open lattice with diagonal bracing to ensure maximum stiffness while reducing weight, and the tapering improves strength and lowers the center of gravity. The rotor and tower were both developed in SolidWorks, and the tower was simulated using FEA subjected to a horizontal force of 1.5 N corresponding to estimated wind forces. Based on the simulation results, the highest stress was found to be 2.555 MPa, the minimum factor of safety was 4.613, and the maximum displacement was 0.1896 mm.

The artificial prototype was subjected to two tests. First, the stiffness of the tower was determined by gradually adding a load from 100 g to 1000 g via a pulley mechanism and noting the deflection caused by this load. The stiffness of the tower was found to be 579.1 N/mm using a linear fit between load and deflection. The turbine model, including its tower, had a total mass of 215.5 g, well within the allowable weight of 300 g. For the second test to determine power generation capabilities, the turbine was mounted in front of a wind source moving at 25 mph, and measurements of voltage, current, and rotor speed were taken.

The highest power generated was 1395 mW (1.395 W) with a current of 595 mA and voltage of 1.93 V, at a speed of approximately 3684 rpm of the rotor. Considering the efficiency of this actual power in comparison with the theoretical power in the wind blowing through the swept area of the rotor, the efficiency of the turbine was approximately 58.9%. This number is close to the Betz theoretical limit, which is a limit that states that no wind turbine can exceed an efficiency of 59%. With this project, our group, the campo-windy, was able to create a lightweight and relatively strong wind turbine that we will analyze more below.

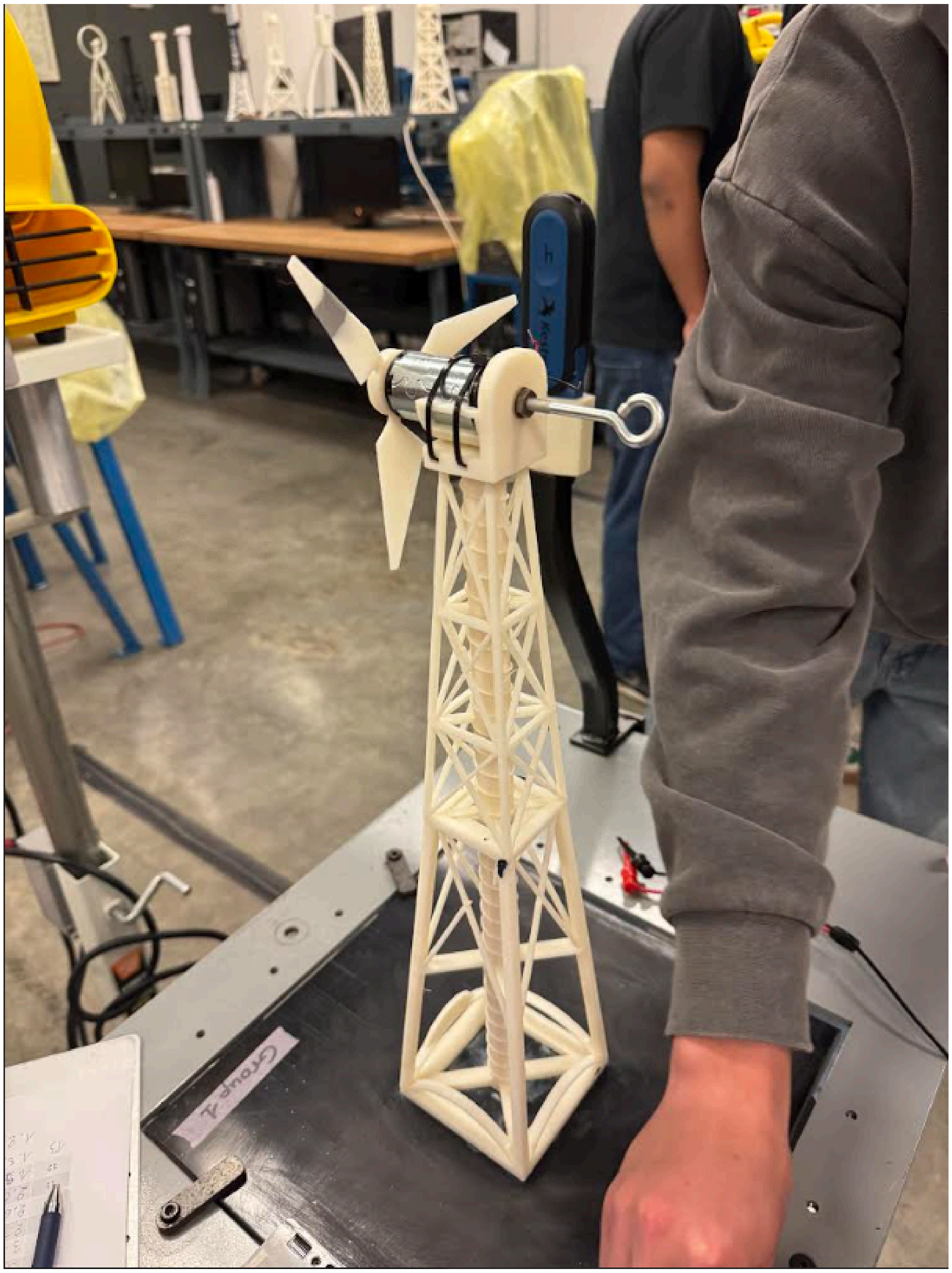


Figure 1. Final Tower with motor and blade mounted

Table of Contents

Introduction

As the global demand for energy increases and the consequences of carbon-based energy become more apparent, it is clear that developing renewable energy sources has become a critical necessity. In particular, wind energy has become one of the most viable and scalable sources of clean energy. Unlike solar plants, wind generation can occur all the time, and modern wind turbines are capable of producing electricity at costs comparable to conventional power plants. In the United States alone, roughly 10% of all energy generated comes from wind energy, and it is the largest renewable source of energy, and that figure is growing every year. Thus, the development and refinement of wind turbine technologies, from small-scale models to large offshore installations, is a very important pursuit with real-world applications.

A wind turbine works by converting the kinetic energy of moving air into rotational kinetic energy, which then drives a generator to produce electricity. When wind hits the turbine blades, the aerodynamic shape of the blades, called an airfoil profile, generates lift. This causes a rotational force on the rotor, which is connected to the generator and produces an electric current. The Betz limit describes the maximum efficiency, or amount of kinetic energy converted to electrical energy, possible for a wind turbine to produce under ideal conditions, and this value is about 59%. The efficiency of this process depends on many different design variables, including but not limited to hub height, the number of blades, the cross-sectional profile of the blades, their angle of attack (angle relative to incoming wind), and angle of twist (angle which the blade twists from root to tip). It is by optimizing all these parameters that the wind turbine becomes the most efficient, and therefore that the greatest energy can be produced.

This project tasked our team with researching, designing, modeling in CAD, and testing our own small-scale horizontal-axis wind turbine to operate under simulated lab conditions of 25 mph wind speeds. Each team member independently researched blade and tower designs before the group decided on one design with certain parameters. The turbine blade was designed around four key parameters as discussed below.

Blade Profile: The blade profile is the cross-sectional shape of the blade. The blade profile is an important factor, as it determines the lift-to-drag ratio of the wind turbine. Many different blade profile candidates were considered, but the group ultimately chose to pursue the NACA 4412 profile, with its ideal life-to-drag ratios at low wind speeds and 12% thickness to chord ratio. Below is a freehand sketch of the blade profile.

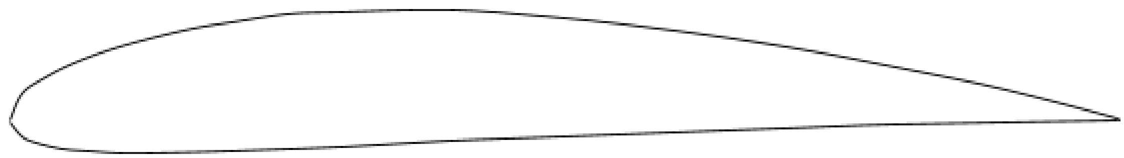


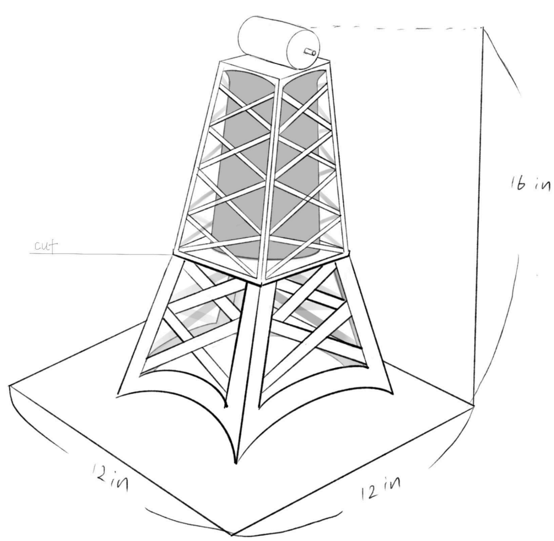
Figure 2. Sketch of wind turbine blade profile.

Number of Blades: This project uses three blades for the wind turbine. Fewer blades produce instability and vibration in the turbine, whereas more blades experience more drag and therefore diminish energy production.

Angle of Attack: The angle of attack describes the angle between the blade chord line and the incoming wind. The ideal angle of attack for a wind turbine varies depending on its blade profile, and ultimately, 6% was chosen as the ideal angle of attack, maximizing the lift-to-drag ratio for the chosen airfoil profile at the given wind speed conditions.

Angle of Twist: The angle of twist refers to the degree of twist from the blade's root to its tip. The angle of twist is another important factor in the efficiency of a wind turbine, but the most ideal number is hard to derive. Eventually, the project team landed on a twist from 20 degrees at the root to 0 degrees at the tip, in line with most other wind turbine designs.

Along with designing the blades, the project also entailed the design of the wind turbine tower. After a few designs, the final product involved a square-shaped truss-based framework to distribute structural loads and provide a better stiffness-to-weight ratio. Below are two sketches that were used to develop the final design.



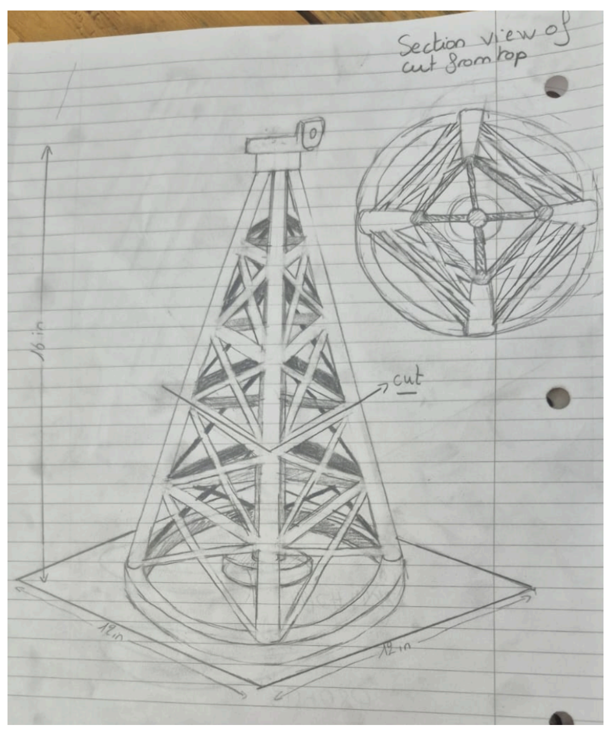


Figure 3 (Top): One preliminary tower design.
Figure 4 (Right): Another preliminary tower design.

Both the tower and the turbine blades were later modeled in CAD on SolidWorks. The team also performed elementary FEA (finite element analysis) to test the rigidity of the tower design.

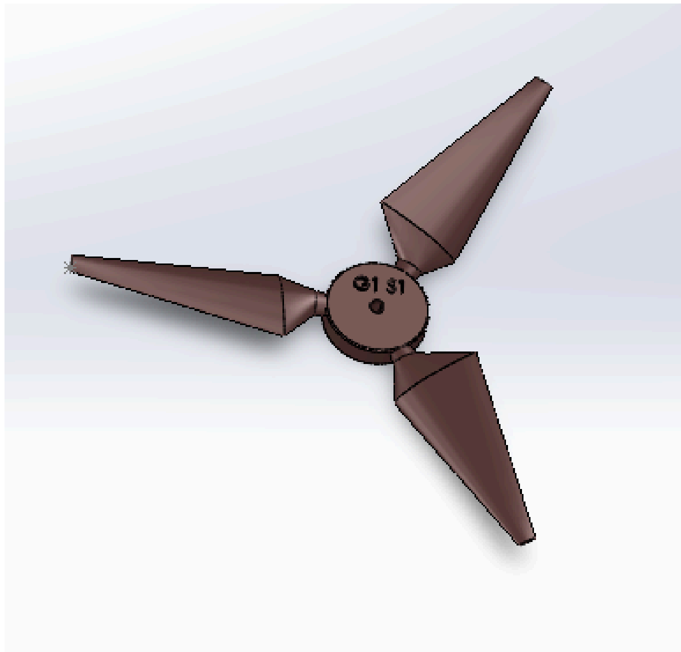


Figure 5. Final blade model on SolidWorks.

The objective of this project was to research and justify blade and tower parameters using published literature, model the turbine through CAD software, validate structural performance through FEA simulation, fabricate the design using 3D printing, and to test the physical prototype in a controlled wind setting to measure power output. The following sections detail the design decisions, fabrication process, analysis, experimental results, and any references and future recommendations.

Wind Turbine Tower and Rotor Blade Design

Design Overview and Final Selection

Our final wind turbine design uses a tower with a tapered truss structure and a three-blade rotor. The tower was designed to support the motor and rotor assembly while minimizing material use and maintaining enough stiffness to resist bending under wind loading. The final tower selection is shown in Figure 5. The structure uses a wide base, vertical support members, diagonal cross-bracing, and a smaller top platform where the motor housing is mounted. This design follows the same general direction as our earlier tower sketches, which emphasized lattice-style support, triangular bracing, and a broad base to improve stability. Our Lab 8 research included several lattice wind turbine tower concepts and references comparing lattice and tubular tower structures, which helped guide the decision to use an open-truss design rather than a solid tower.

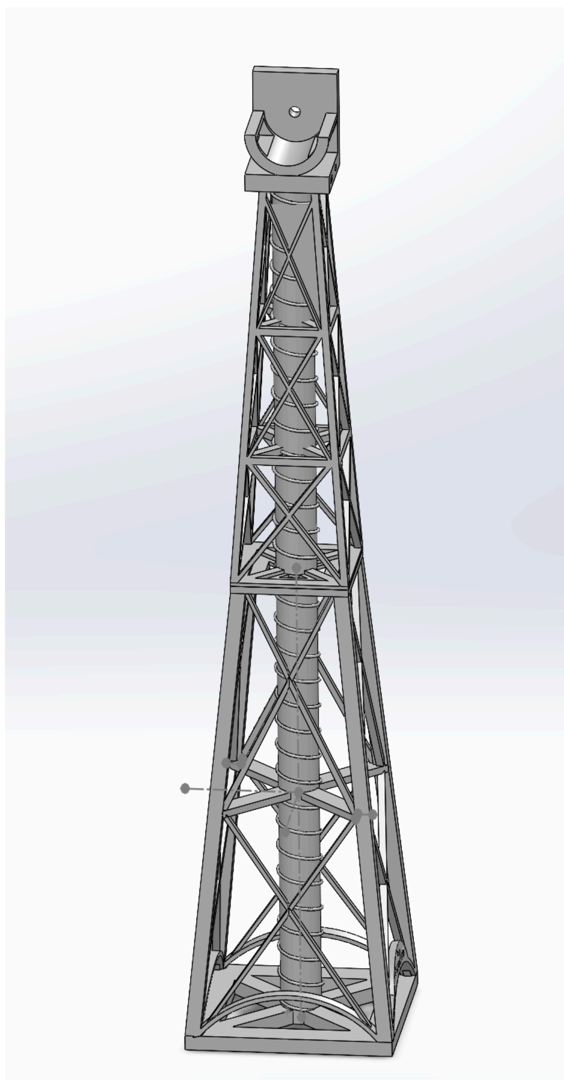


Figure 6. Final selected tower design

The final tower uses repeated triangular sections because triangles are structurally stable and resist deformation better than rectangular frames without diagonal bracing. The tower tapers from a larger base to a smaller top, which helps lower the center of gravity and provides a stronger foundation against overturning. The diagonal lattice members distribute bending loads through multiple paths rather than relying on a single solid column. This was important because the tower needed to be stiff enough to hold the rotor steady while remaining practical for 3D printing.

The rotor blade design was based on our final selection for Lab 5 (Figure 4). The selected blade design uses three blades, a NACA 4412 airfoil profile (Figure 1), an angle of attack of 6°, and a blade twist from 20° at the root to 0° at the tip. The final dimensions from our selection were: blade length of 2.75 in, blade starting position 0.25 in from the center of the hub, maximum profile length of 0.90 in, tip profile length of 0.25 in, maximum profile thickness of 0.240 in, minimum profile thickness of 0.065 in, and root diameter of 0.22 in. The final design also required the blade thickness to be greater than 1/16 in and included a fillet at the end for smoother geometry and improved printability.

The three-blade rotor was selected because it provides a balance between torque, smooth rotation, and structural stability. Lab 5 research showed that three-blade designs are a strong practical choice because they reduce vibration and torque ripple compared with one- or two-blade designs, while avoiding the additional drag and material usage associated with four or more blades. The NACA 4412 profile was selected because it provides a useful balance between aerodynamic performance and structural thickness. Since the blade is 3D printed, the 12% thickness of the NACA 4412 profile helps simplify manufacturing while maintaining aerodynamic performance.

Design Tool

The tool used to design the wind turbine was SolidWorks. SolidWorks was used to create the 3D CAD model of the tower, motor mount, and blade geometry. The tower was modeled using vertical members, diagonal truss members, and a base support. The rotor blades were modeled using the selected airfoil profile and dimensions from the final design. SolidWorks was also used to check the model's physical properties, including the tower's volume, and to run a Finite Element Analysis simulation to test the tower's stiffness under applied loads.

Finite Element Analysis Simulation

A Finite Element Analysis simulation was performed to determine the deflection, stiffness, and structural safety of the wind turbine tower. The purpose of the simulation was to predict how much the tower would bend when a horizontal load was applied near the top, where the motor and rotor are mounted. This loading condition represents the force created by wind pushing on the rotor and motor assembly during testing.

For this simulation, the bottom of the tower was fixed to represent the tower being attached to the testing platform. A horizontal force of 1.5 N was applied at the top of the tower near the motor mount. This force was chosen as a realistic test load because the estimated wind load from the 25 mph fan is expected to be around the 1–2 N range. The simulation used ABS as the material and generated three main result plots: von Mises stress, factor of safety, and total displacement.

Stress Results

The von Mises stress plot showed that most of the tower experienced very low stress, shown by the mostly dark blue color across the lattice structure. The maximum von Mises stress was:

$$\sigma_{max} = 2.555 \times 10^{6} N/m^{2}$$

The highest stress occurred near the top motor mount, where the applied force was applied. This result makes sense because the motor mount transfers the horizontal load into the tower frame. The lower tower and most of the diagonal bracing remained blue, indicating the stresses in those areas were much smaller than the maximum.

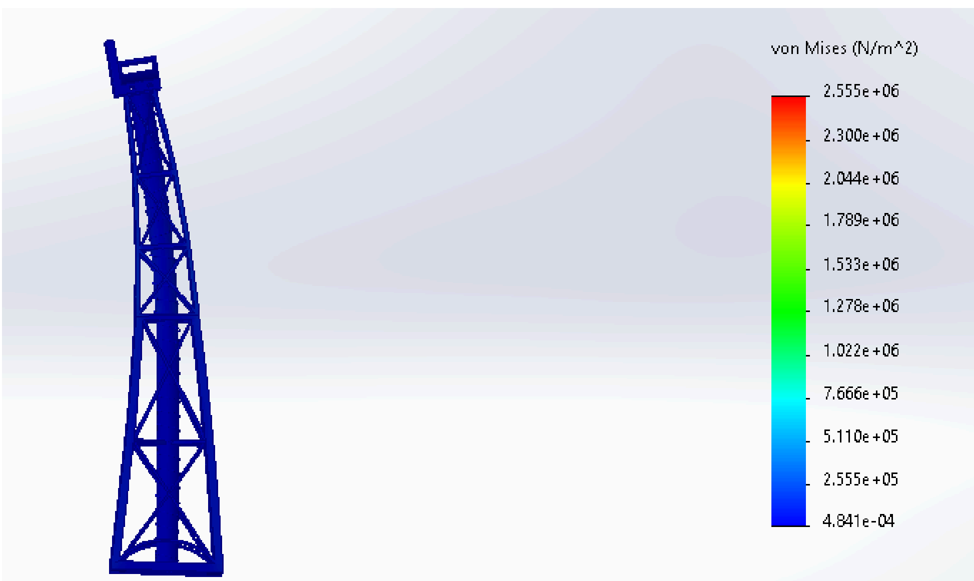


Figure 7. Von Mises stress plot under 1.5N horizontal load

Maximum stress: $2.555 \times 10^{6} N/m^{2} = 2.555 MPa$

Factor of Safety Results

The factor of safety plot showed a minimum factor of safety of:

$$FOS_{min} = 4.613$$

This means that, under the 1.5 N applied load, the tower is predicted to withstand more than four times the simulated load before failure, according to the SolidWorks material model. Since the minimum factor of safety exceeds 1, the design is considered structurally safe for this load case.

The plot also shows very large factor of safety values in some low-stress regions, with the upper scale reaching approximately:

$$6.404 \times 10^{10}$$

This large range causes many of the load-carrying members to appear red even though they are not failing. In this plot, red represents the lowest safety factor, not necessarily failure. Because the minimum factor of safety is still 4.613, the tower remains safe under the 1.5 N load.

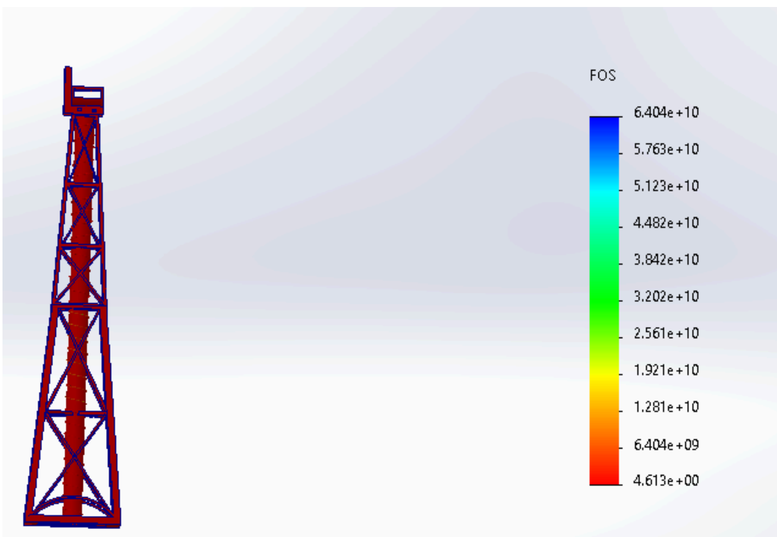


Figure 8. Factor of safety plot under 1.5N horizontal load

Minimum factor of safety: 4.613

Displacement Results

The displacement plot measured the total resultant displacement. The maximum displacement was:

$$\delta_{max} = 1.896 \times 10^{-1}\, mm$$

The largest displacement occurred at the top of the tower near the motor mount, while the base showed nearly zero displacement because it was fixed. This behavior is expected because the tower bends from the bottom support, and the top has the greatest freedom to move. The displacement is small relative to the overall tower height, indicating that the lattice structure is relatively stiff under the 1.5 N horizontal load.

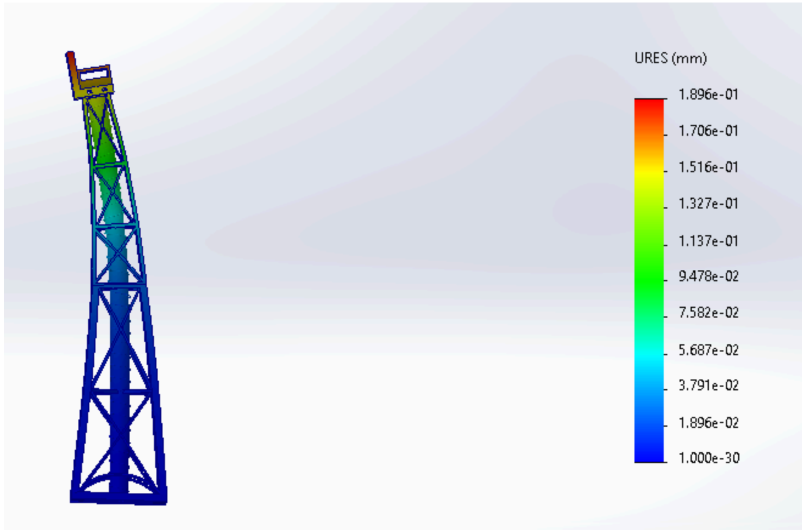


Figure 9. Total displacement plot under 1.5N horizontal load
Maximum displacement: 0.1896 mm

Load vs. Deflection Data

To evaluate the tower's stiffness, multiple SolidWorks FEA simulations were run with varying horizontal loads applied at the top motor mount. For each load case, the maximum resultant displacement, or URES, was recorded from the displacement plot. The base of the tower was fixed for every simulation, and the load was applied in the horizontal wind direction.

FEA displacement was recorded from 0.5N to 6N with increments of 0.5. The graph was arranged as:

$$x = \delta = deflection\ in\ mm$$
$$y = F = applied\ load\ in\ N$$

This format is used because the slope of the line directly represents the stiffness of the tower:

$$F = k\delta$$

where k is the tower stiffness.

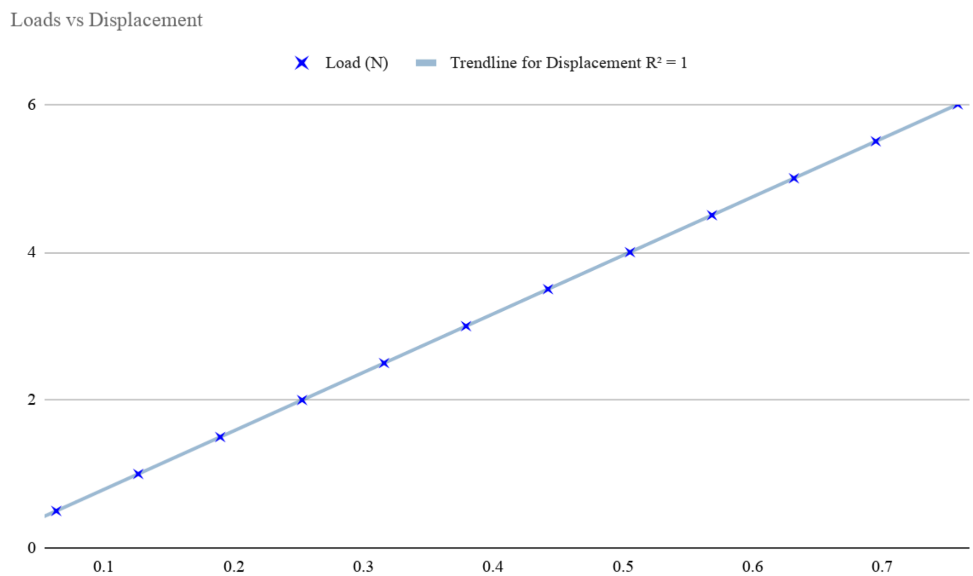


Figure 10. Loads vs. Deflection plot

A linear trendline was added to the load-deflection graph. The resulting trendline equation was:

$$F = 7.9134\delta - 0.000093$$

where:

$$F = applied\ load\ in\ N$$
$$\delta = maximum\ deflection\ in\ mm$$

The correlation factor was:

$$R^2 = 0.9999999814$$

The R^2 value is extremely close to 1, which shows that the relationship between load and deflection is almost perfectly linear. This means the tower behaved elastically within the tested loading range. As the load increased, the deflection increased proportionally.

The slope of the line represents the stiffness of the tower is $k = 7.9134\ N/mm$. This means that approximately 7.91 N of horizontal force is required to create 1 mm of deflection at the top of the tower. Since the maximum deflection under the main 1.5N load was only $\delta = 0.1896\ mm$, the tower can be considered relatively stiff under the expected wind loading conditions.

Tower Volume

The volume of the final wind turbine tower was obtained using the Mass Properties tool.

$$V = 14.12\ in^3$$

This volume shows that the tower uses relatively little material while still maintaining good stiffness. The lattice-style design reduces the amount of 3D-printed material compared with a solid tower, while the diagonal cross-bracing helps resist bending. The wide base provides stability, and the tapered upper structure reduces unnecessary material near the top of the tower.

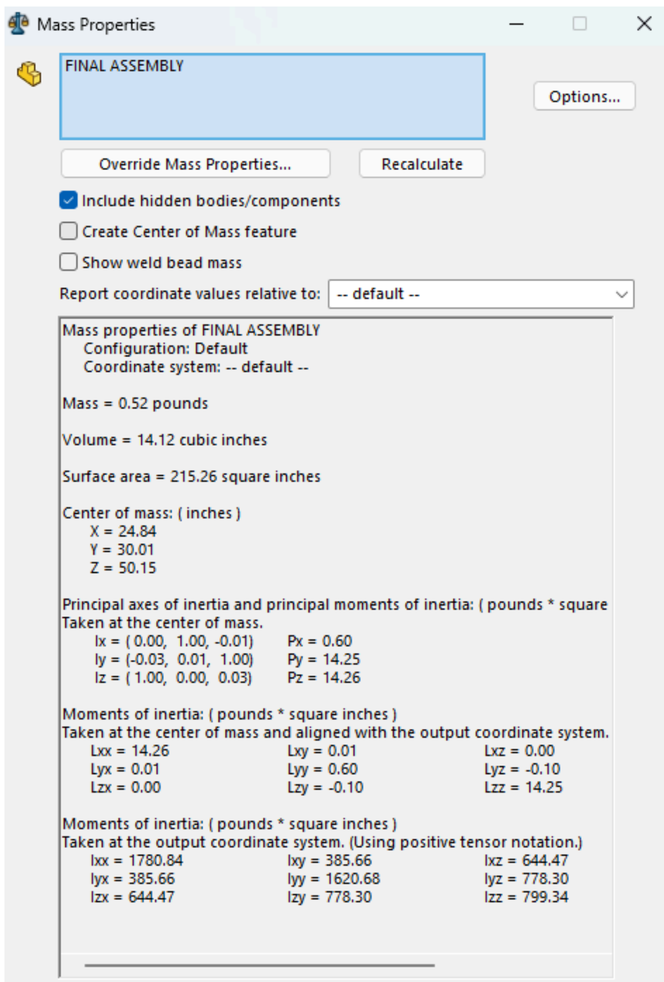


Figure 11. Tower Volume

Testing and Results

Once we had completed the concepting, design, and construction of the wind turbine, it was time to test our tower. The wind turbine was tested to evaluate the **structural stiffness** and **electrical power generation**.

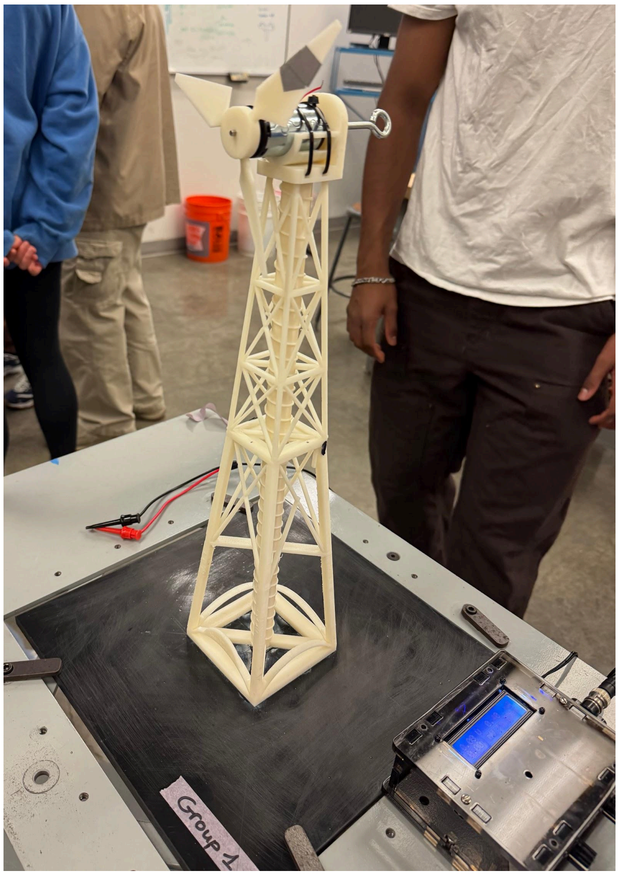


Figure 12. Wind turbine power-generation testing setup with a DC electronic load and a digital power meter.

For the power generation test, the turbine was placed in front of the wind source fan and tested at a wind speed of 25 mph at the rotor location. The turbine was secured and equipped with a small motor mounted on top of the tower, holding the blades. The turbine output was connected to a DC electronic load and a digital power meter, which measured the turbine's voltage, current, and electrical power. To better understand the turbine's capabilities, we tested it under various electrical loads. For each collected data point, the electrical load was slightly adjusted, and the corresponding voltage, current, power, and blade rotational speed were recorded and listed in Table 1 listed below.

For the tower stiffness test, incremental loads of 100 grams were added on to the turbine tower, and the resulting lateral displacement was measured in millimeters and recorded in Table 2. This allowed our team to determine the relationship between applied load and displacement. The stiffness of the tower can be found from the slope of the load-displacement plot.

Power Generation Results

The power generation data collected during testing is shown in Table 1 below. As the current increased, the voltage generally decreased, which is expected because a greater electrical load draws more current from the generator and causes a drop in output voltage. The power output increased over most of the testing range before reaching a maximum value at data point 12.

Table 1. Wind turbine power generation data at 25 mph wind speed.

Data Point	Voltage (V)	Current (mA)	Power (mW)	Blade Rotational Speed (rpm)
1	3.86	73	280	5478
2	3.73	104	375	5258
3	3.50	139	480	5354
4	3.40	172	598	5301
5	3.30	204	680	5271
6	3.18	241	770	5056
7	3.05	290	880	4916
8	3.01	319	970	4897
9	2.78	375	1030	4709
10	2.60	435	1140	4492
11	2.01	575	1283	3871
12	1.93	595	1395	3684
13	1.82	653	1170	3622
14	1.26	670	860	3024

The maximum recorded experimental power was approximately <u>1395 mW</u>, or <u>1.395 W</u>, at a current of <u>585 mA</u> and a voltage of <u>1.83 V</u>. This indicates that the turbine reached its peak measured power output before power began to decrease at higher current values. This behavior suggests that after a certain point, increasing the electrical load caused the generator voltage and rotor speed to overload, resulting in a decrease in total power output.

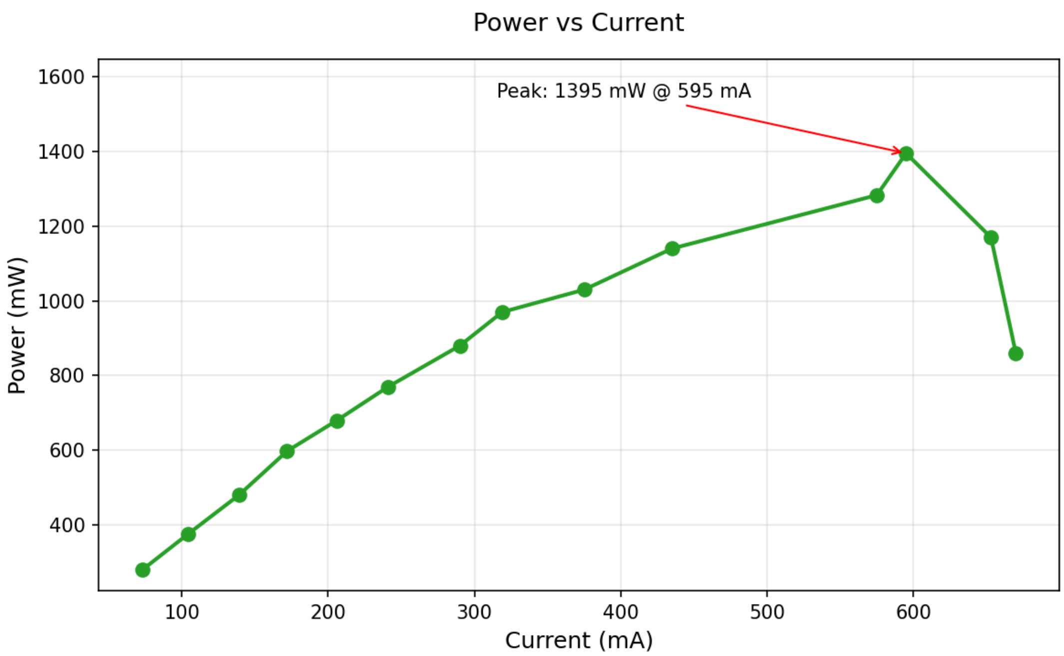


Figure 13. Power vs. Current for the wind turbine at 25 mph.

The **voltage** vs. **current** plot shows a decreasing trend. This means that as the turbine was required to supply more current, the voltage dropped. This is expected for a small generator because higher electrical loading increases the resistance experienced by the rotor which would reduce rotational speed and output voltage.

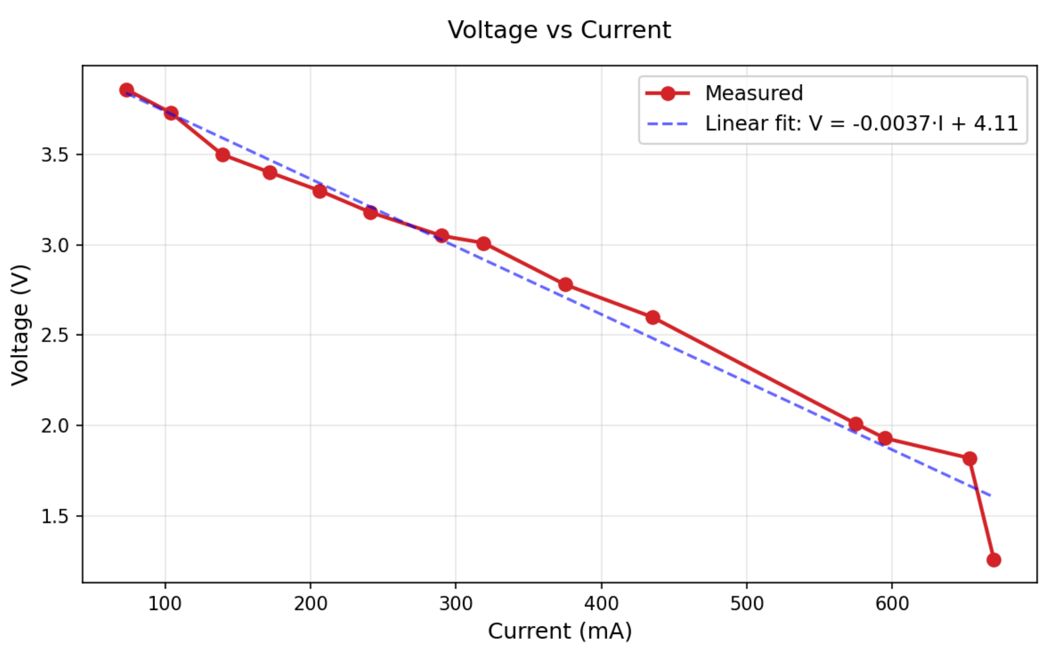


Figure 14. Voltage vs. Current for the wind turbine at 25 mph.

Tower Stiffness Results

As explained earlier, the tower's stiffness was tested through hanging weights off a pulley on the side and pulling the top of the tower. The tower stiffness data is shown in Table 2. The total weight of the turbine was recorded as **215.5 g**. During testing, the applied load was increased, and the resulting displacement was measured.

Table 2. Tower stiffness test data.

Data Point	Load (grams)	Displacement (mm)
1	100	0.15
2	200	0.44
3	300	0.55
4	400	0.78
5	500	0.79
6	600	1.15
7	700	1.21
8	800	1.48
9	900	1.58
10	1000	1.75

Based on the trend of Figure D, the tower behaved mostly linearly over the tested load range. In other words, displacement increased predictably as load increased. A stiffer tower is desirable because too much deflection can reduce turbine stability, affect interior rotor alignment, and lower power generation efficiency. With efficiency and safety concerns, a stiff tower is necessary.

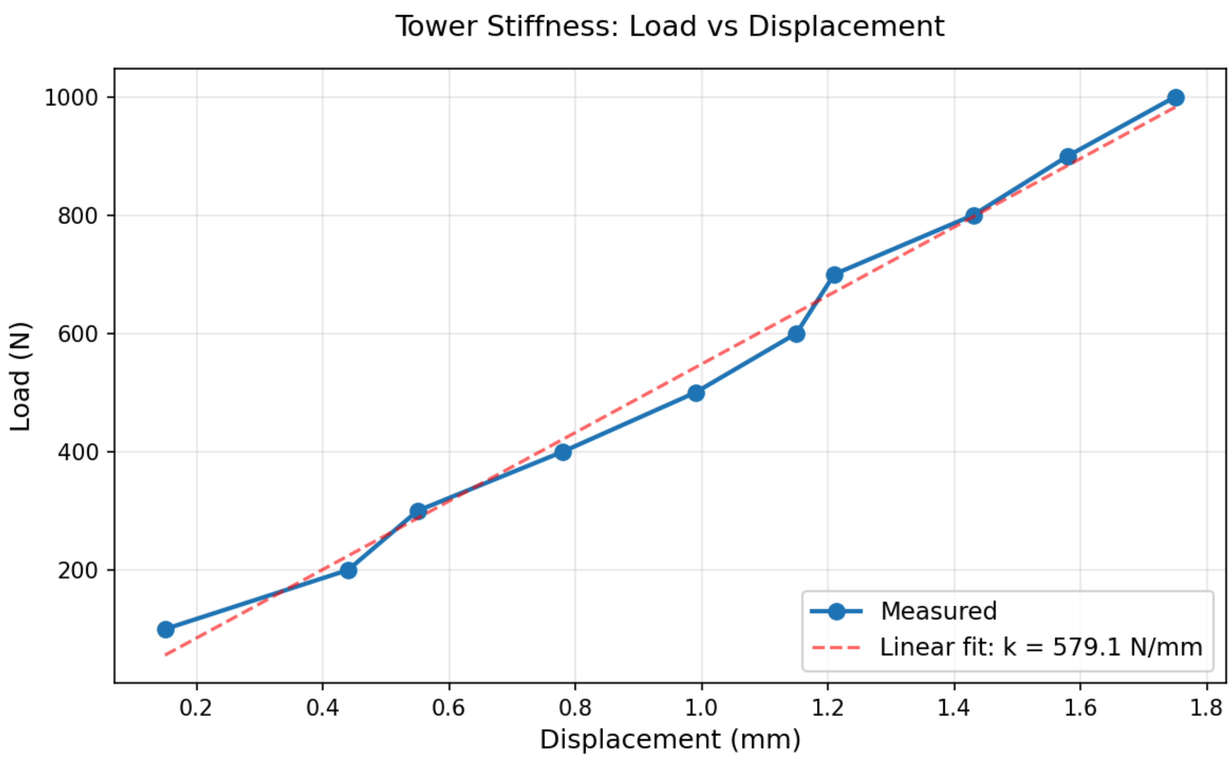


Figure 15. Load vs. Displacement for the wind turbine tower.

Efficiency Calculation

The efficiency of the wind turbine can be calculated by comparing the experimental power output to the theoretical wind power available at the rotor. The efficiency is 58.9% with the given calculation of Experimental Power divided by Theoretical power multiplied by 100.

Efficiency Calculation:

$$\text{Efficiency}(\eta) = \frac{P_{experimental}}{P_{theoretical}} \times 100$$

$\rho = 1.225\,kg/m^3$ air density

$A = \pi(0.03\,m)^2 = 0.002827\,m^2$

$V = 25\,mph = 11.18\,m/s$

$$P_{theoretical} = \frac{1}{2}\rho A v^3 = \frac{1}{2}\cdot\left(1.225\,\frac{kg}{m^3}\right)\cdot(0.002827\,m^2)\cdot(11.18\,m/s)^3 \approx 2.368\,W$$

$P_{experimental} = 1395\,mW = 1.395\,W$

$$\eta = \frac{1.395}{2.368} \times 100 = 58.9\,\%$$

CAD Drawings

3D assembly drawing of the wind turbine, exploded view of the tower. Also, include a 2D drawing the tower with over all dimensions.

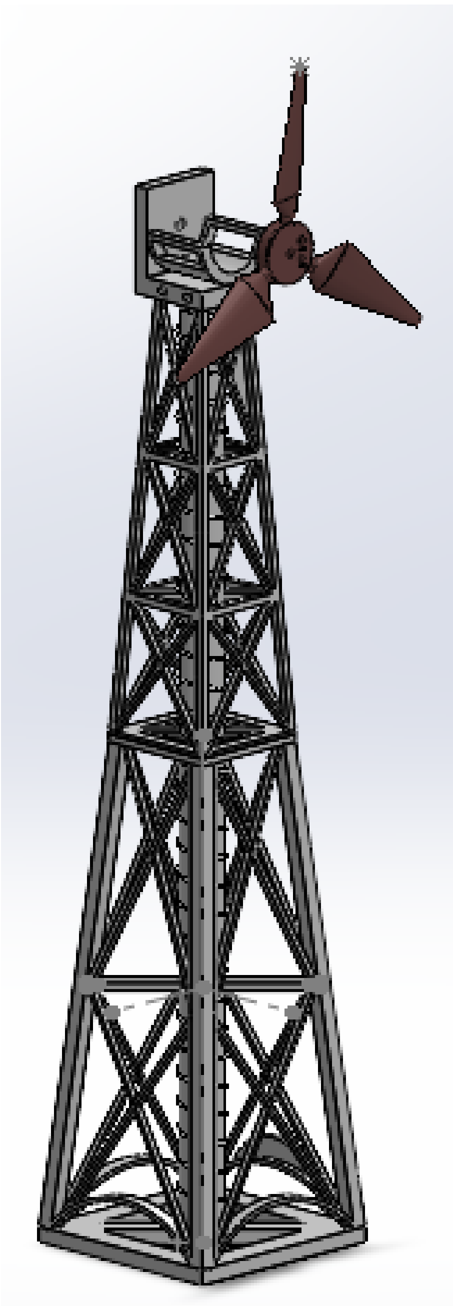


Figure 16 (left). 3D Assembly Drawing of the Wind Turbine

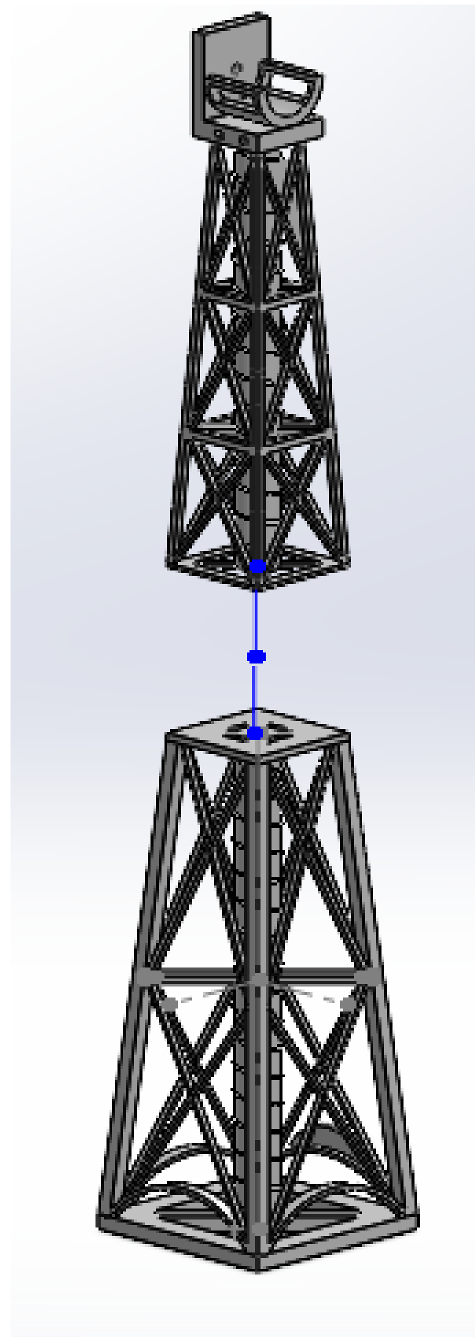


Figure 17 (right). Exploded View of the Tower

Figure 18. 2D Drawing of the tower

Conclusions

Overall, this project successfully produced a functional small-scale horizontal-axis wind turbine that combined aerodynamic blade design, structural tower design, SolidWorks modeling, FEA validation, 3D printing, and physical testing. The final design used a three-blade rotor with a NACA 4412 airfoil, a 6° angle of attack, and a blade twist from 20° at the root to 0° at the tip. The tower used a tapered square-truss structure with diagonal cross-bracing, which kept the design lightweight while providing sufficient stiffness to support the rotor and motor assembly during testing.

The FEA results showed that the tower was structurally safe under the simulated wind loading condition. Under a 1.5 N horizontal load applied at the top motor mount, the maximum von Mises stress was 2.555 MPa, the minimum factor of safety was 4.613, and the maximum displacement was only 0.1896 mm. These results indicate that the tower was not close to failure and that the greatest deflection occurred where expected: near the top motor mount, where the horizontal wind load was applied.

The load-deflection results further confirmed that the tower behaved elastically within the tested FEA range. The FEA load-deflection plot produced the equation $F = 7.91348 - 0.000093$ with $R^2 = 0.9999999814$. This near-perfect R^2 value shows that deflection increased almost exactly proportionally with load. The slope of the line gave a tower stiffness of approximately 7.9134 N/mm, meaning the tower was relatively stiff under the expected wind loading conditions. The SolidWorks Mass Properties tool also showed that the final tower volume was only 14.12 in^3, supporting the idea that the lattice structure was material-efficient while still maintaining structural performance.

Physical testing showed that the turbine generated measurable electrical power at the required 25 mph wind speed. The maximum recorded power output was 1.395 W at approximately 595 mA, 1.93 V, and 3684 rpm. The turbine's calculated efficiency was approximately 58.9%, close to the theoretical Betz limit of 59%. The voltage-current and power-current trends also showed the expected generator behavior: as current increased, voltage decreased, and power increased until reaching a peak, then dropped under heavier electrical loading.

The final wind turbine met the project's main goals by integrating research-based design, CAD modeling, FEA simulation, 3D printing, and experimental testing into a single working prototype. The turbine was structurally stable, relatively lightweight, and capable of generating electrical power in the test environment.

Recommendations for Future Work

Based on the results of this project, the following recommendations show specific changes that would improve the turbine's performance, structural integrity, and design process in potential future iterations.

The most important improvement would be increasing the blade radius. Our rotor swept area was based on a blade length of only 2.75 inches, and since theoretical power scales with the square of the rotor radius, even a modest increase, say from 2.75 to 4 inches, would significantly raise the power maximum. This matters because our turbine only achieved 58.9% efficiency, meaning a significant portion of the available wind energy went uncaptured, and a larger swept area is one of the most direct ways to make it more efficient. With this, the angle of attack of 6° and twist from 20° to 0° were chosen based on general guidelines but were never experimentally optimized. A good way to improve could be to test multiple angles of attack settings at the same wind speed and compare power output directly to find a better operating point for this specific generator and blade geometry.

The sharp power drop-off after data point 12 (595 mA, 1.93 V) indicates the turbine briefly hit its peak and then became overloaded quickly, which means the turbine was only operating at its most efficient point for a very narrow range of conditions. This suggests a mismatch between the generator's internal resistance and the external load being applied. A simple solution would be to calculate the generator's internal resistance from the voltage-current data, using the linear fit equation $V = -0.0037 \cdot I + 4.11$, and then select a fixed load resistor matched to that value. This would allow the turbine to operate closer to its maximum power point for the entire test rather than briefly passing through it.

For the tower, our structure weighed 215.5 grams against a 300 gram allowance, leaving nearly 85 grams of unused weight. This is important because a heavier, more material-dense structure would have been stiffer and more resistant to lateral deflection, which directly affects rotor alignment and power generation consistency. Rather than simply adding bulk, we would specifically thicken the diagonal bracing members in the lower half of the tower, since the FEA stress plots confirmed that region experienced the highest bending stress under lateral load. Increasing member thickness there would give the most stiffness improvement per gram of added material, and would bring the design closer to fully utilizing the weight constraint.

The print quality issues came directly from some tower members being too thin for the SLA resin printer to reproduce reliably, which resulted in sections of the tower being weaker and more fragile than what was modeled in SolidWorks. This likely affected both the stiffness test results and the overall structural integrity of the tower during testing. SLA printers generally require a minimum wall thickness of around 1–2 mm for structural parts, so before the next print, every member in the SolidWorks model should be checked to be thicker than that and anything below it should be thickened or consolidated with adjacent members. It would also be worth considering switching to FDM printing for the tower, since FDM produces parts that are generally tougher and more resistant to mechanical loading, which is more useful for a structural component like this than the smooth surface finish that SLA provides.

Finally, the FEA simulation results and the physical stiffness test results showed a large discrepancy that should be investigated in future iterations. The FEA predicted a tower stiffness of 7.9134 N/mm, while the physical stiffness test measured 579.1 N/mm, a difference of nearly two orders of magnitude. This gap most likely comes from the FEA material properties not accurately reflecting the actual behavior of the SLA-printed resin, or from differences between how the boundary conditions were set up in SolidWorks and how the tower was actually loaded and clamped during testing. In future iterations, the FEA should be calibrated against the physical results early in the process, and the simulation inputs should be adjusted until the predicted and measured stiffness values are reasonably close. This would make the FEA a genuinely useful design and prediction tool rather than just a required step.

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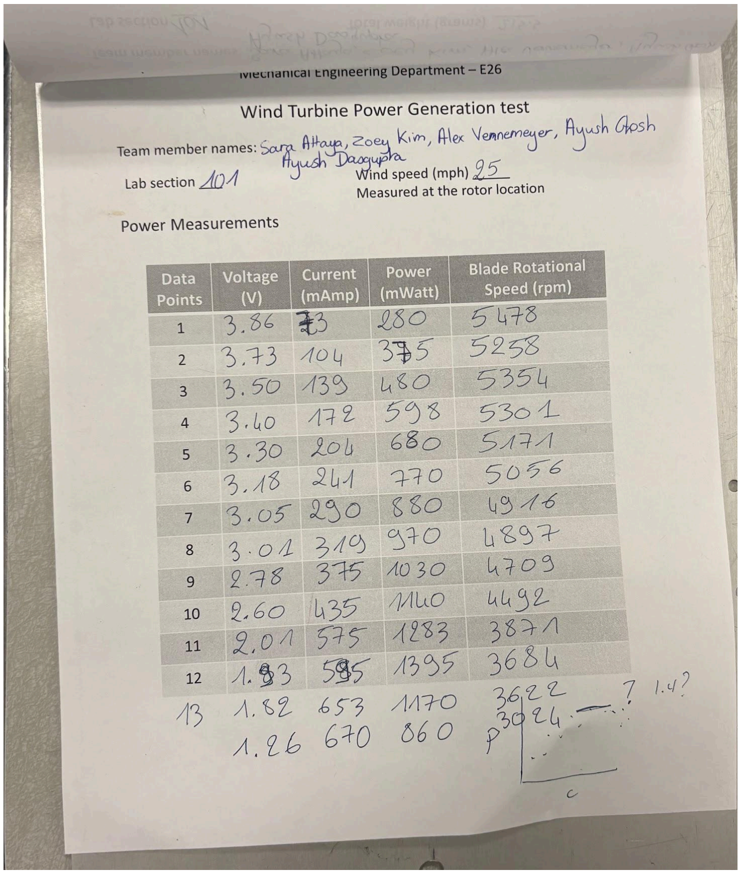
Appendix

While references cover all our citations, here are the recording sheets:



Mechanical Engineering Department – E26

Wind Turbine Power Generation test

Team member names: Sara Attaya, Zoey Kim, Alex Vennemeyer, Ayush Ghosh, Ayush Dasgupta

Lab section 101

Wind speed (mph) 25
Measured at the rotor location

Power Measurements

Data Points	Voltage (V)	Current (mAmp)	Power (mWatt)	Blade Rotational Speed (rpm)
1	3.86	73	280	5478
2	3.73	104	375	5258
3	3.50	139	480	5354
4	3.40	172	598	5301
5	3.30	204	680	5171
6	3.18	241	770	5056
7	3.05	290	880	4916
8	3.01	319	970	4897
9	2.78	375	1030	4709
10	2.60	435	1140	4492
11	2.01	575	1283	3871
12	1.83	595	1395	3684
13	1.82	653	1170	3622
	1.26	670	860	3024

? 1.4?

P

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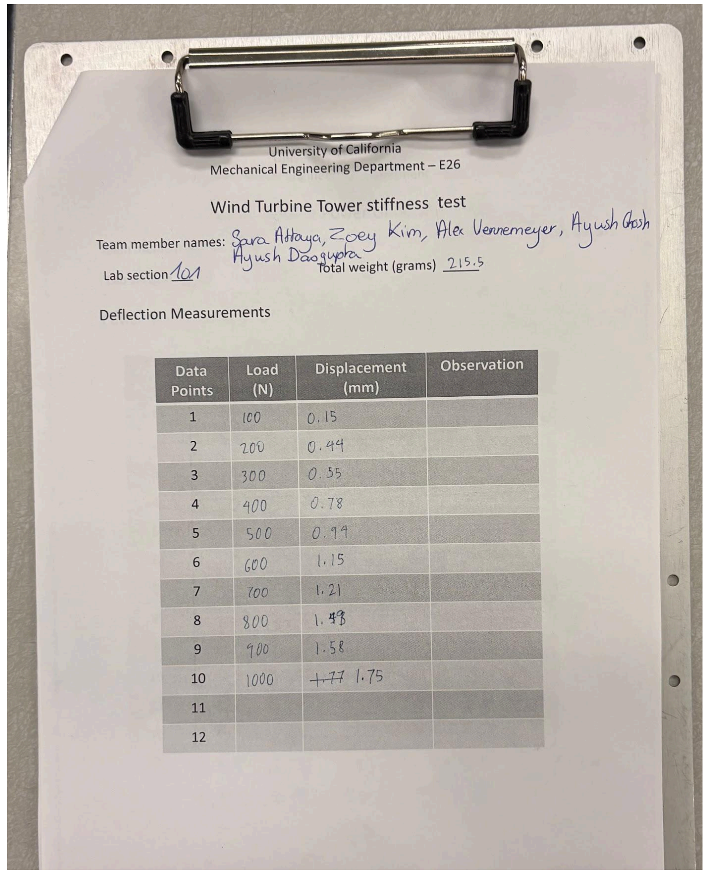

University of California
Mechanical Engineering Department – E26

Wind Turbine Tower stiffness test

Team member names: Sara Attaya, Zoey Kim, Alex Vennemeyer, Ayush Ghosh, Ayush Dasgupta

Lab section 101

Total weight (grams) 215.5

Deflection Measurements

Data Points	Load (N)	Displacement (mm)	Observation
1	100	0.15	
2	200	0.44	
3	300	0.55	
4	400	0.78	
5	500	0.94	
6	600	1.15	
7	700	1.21	
8	800	1.48	
9	900	1.58	
10	1000	~~1.77~~ 1.75	
11			
12			